

SMART INDIA HACKATHON 2026

BlockLedger

Blockchain-Based Secure Platform for Identity,
Access Control and Digital Asset Management

PS ID

26125

ORGANISATION

Bharat Electronics Limited

CATEGORY

Software

THEME

**Blockchain &
Cybersecurity**

Team Mercury · Netaji Subhas University of Technology

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Centralised registers cannot prove their own integrity

01

Single point of failure

One compromised directory administrator can mint, escalate or delete any identity. There is no independent record, because the same administrator controls the log.

02

Identity theft and impersonation

Credentials are bearer secrets held centrally. Whoever obtains the store obtains every identity in it. Employees hold no portable, provable identity.

03

Unverifiable asset provenance

A drawing, firmware image or certificate can be modified after approval, and a downstream consumer has no cryptographic way to prove which bytes were approved.

Net effect: an auditor cannot independently prove that a log has not been altered after the fact.

Replace the trusted record with cryptographic evidence

Decentralised identity

Every actor is a DID — `did:blkl:sol:<pubkey>` — with a W3C DID Document controlled by a keypair, not a directory row. Pinned to IPFS, so it is content-addressed.

NFT-based asset ownership

Every asset is a token (BLKL-000123) whose metadata commits to a SHA-256 digest of the file. Change one byte and the digest no longer matches.

Contract-enforced access control

Privileged actions pass a permission check that reverts with an explicit reason. A caller cannot bypass a check by ignoring a return value.

Tamper-evident audit trail

Each entry commits to the hash of its predecessor, so retroactive edits are detectable in a single pass — and anchored to Solana devnet.

How a request flows through the system

User + Solana Wallet

Phantom / Solflare

Next.js Frontend — App Router, React 19

dashboard · identity · assets · access-control · audit · wallet

Context Layer

Auth · Chain · Ledger · Storage

Contract Layer

AccessControl · IdentityRegistry · AssetNFT — revert on failure

Blockchain Core

DID derivation · SHA-256 hash chaining · append-only ledger + verifier

IPFS · PINATA

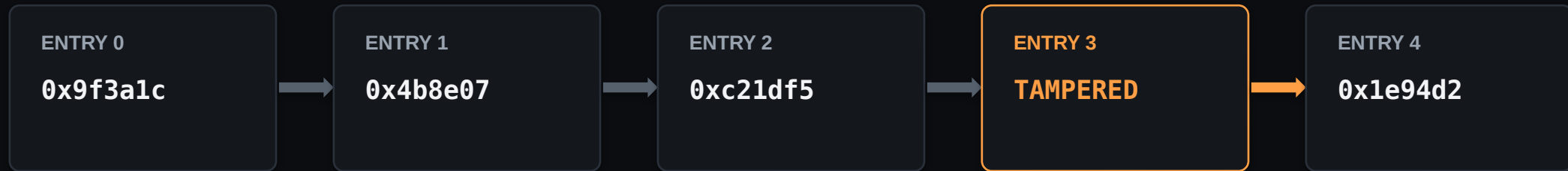
DID Documents and NFT metadata
pinned to real, content-addressed CIDs.

SOLANA DEVNET

Audit entries anchored as real signed
memo transactions, linked to an explorer.

A log that proves it has not been edited

```
entry[n].hash = SHA-256( entry[n].payload || entry[n-1].hash )
```



```
verifyChain() → { valid: false, brokenAt: 3 }
```

Four classes of tampering are detected, each reporting the exact broken index:



All four are demonstrated live in the prototype.

KEY FEATURES

What the prototype does today

Identity

Register, resolve, revoke and reactivate DIDs with W3C DID Documents pinned to IPFS.

Access control

Four roles mapped through an explicit permission matrix, enforced in UI and contract layer alike.

Real reverts

Unauthorised calls fail loudly with Solidity-style errors, surfaced directly to the user.

Digital assets

Mint DID-bound NFTs with content digests, IPFS metadata and full transfer history.

Audit trail

Hash-chained, append-only log with one-click verification and export.

On-chain anchor

Signed Solana devnet memo transactions, verifiable in a public block explorer.

Built on proven, open components

FRONTEND

- Next.js 16 — App Router
- React 19
- TypeScript
- Tailwind CSS v4
- shadcn/ui
- Framer Motion

BLOCKCHAIN

- Solana — devnet
- @solana/web3.js
- Wallet Adapter
- W3C DID
- SHA-256 hash chaining
- Web Crypto API

STORAGE & DEPLOY

- IPFS
- Pinata pinning
- Content-addressed CIDs
- Vercel
- npm
- Zero contract deployment

What is real, and what is simulated

We state this explicitly rather than overclaim. Nothing in the demo is fabricated.

REAL

- Solana devnet wallet connection
- Signed memo transactions anchoring audit entries
- IPFS pinning via Pinata — genuine CIDs
- DID derivation and W3C DID Documents
- Hash chaining and chain verification

SIMULATED

- Contract layer written in TypeScript, not deployed
- Enforcement logic and error semantics mirror an on-chain program
- Demo authentication uses seeded credentials
- Chosen so the prototype runs with zero deployment cost

Every record carries `anchorMode: "onchain" | "simulated"` and is badged truthfully in the UI. No signature or CID is ever fabricated.

Five steps a reviewer can reproduce

- 1 ADMIN registers an identity** */identity — a DID is issued and its DID Document is pinned to IPFS.*
- 2 MANAGER mints a digital asset** */assets — metadata pinned, token bound to an owner DID, digest recorded.*
- 3 AUDITOR is denied a role grant** */access-control — a real contract revert is raised and surfaced.*
- 4 Chain is verified, then tampered** */audit — the verifier reports the exact index at which the chain breaks.*
- 5 Entry is anchored on-chain** */wallet — a signed devnet memo transaction, viewable in a block explorer.*

From prototype to deployment

IMPACT

- Removes the administrator as a single point of trust
- Gives auditors mathematical proof of log integrity
- Makes asset provenance verifiable by any party
- Portable identity that survives directory migration
- Independent verification without vendor cooperation

FUTURE SCOPE

- Deploy the contract layer as Solana programs via Anchor
- Wallet-native, signature-based authentication
- Verifiable Credentials for cross-organisation attestation
- Hardware wallet and HSM-backed keys for ADMIN accounts
- Merkle-root batch anchoring to cut cost at scale
- Phased LDAP / Active Directory integration

BlockLedger

Identity you own. Access you can prove. Assets you can trace.

TEAM MERCURY

Harsh (Team Leader) · Anushka · Sanyyam · Prakhar · Zoya · Shauryan

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github.com/hxrshxz/BlockLedger